

Retinex-Based Hybrid Framework For Low-Light Image Enhancement

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Abstract -

This paper talks about a new hybrid method for improving low-light images. Basically, it mixes deep learning with real-time image enhancement techniques. Two modes are there. And they work together pretty well. The Backend AI Mode uses transformer architectures, along with Retinex theory and diffusion models, to give high-quality results. It's more focused on quality. On the other hand, the Frontend Default Mode uses optimized computer vision algorithms. This one is fast. Like really quick processing. So yeah, this hybrid setup lets users choose. Either go for better quality or faster speed, depending on what they need. We also tried something a bit different. A unique way of combining transformer-based feature extraction with physics-inspired Retinex decomposition and diffusion refinement. Sounds complex, but it works. And honestly, it performs better than many existing methods. When we tested it using the LOL dataset, the Backend AI Mode showed better PSNR and SSIM scores. And no major increase in processing time, which is good. Meanwhile, the Frontend Default Mode gives acceptable quality, with processing speeds under one second. So it's quite useful for interactive stuff. ser studies also suggest that the system is helpful in different low-light situations. Not perfect in every case, but yeah, works well overall.

Index Terms — low-light enhancement, image processing, deep learning, transformer, diffusion models, real-time processing, Retinex theory, web application

I. INTRODUCTION

Low-light image enhancement is actually a pretty common problem in computer vision. When images are taken in low light, they usually don't look good. The visibility is low, contrast is poor, colors look off, and there's a lot of noise too. So yeah, not very useful in many cases.

There are some traditional methods to fix this. Things like histogram equalization, gamma correction, and unsharp masking are used. They can improve brightness and contrast to some extent. But honestly, they are not perfect. Sometimes they create weird artifacts or remove fine details. Physics-based methods like Retinex theory try to solve this in a better way by modeling how images are formed. CNNs are commonly used and give good results. More recently, transformer-based models are also being used because they can capture global information better. But the problem is they require high computation power. So using them in real-time or in simple systems becomes difficult.

So in this paper, we tried to handle this issue in a different way. we built a hybrid system. It has two modes. First one is the Backend AI Mode which focuses more on quality and uses transformers with Retinex theory and diffusion models to get better results. It is a bit slower but quality is good.

The second one is the Frontend Default Mode. This is designed for speed and runs directly in the browser and uses optimized computer vision techniques. The quality is not as high as the backend mode, but it's fast enough for real-time use, which is important in many applications.

The rest of the paper is organized like this. First, Section II talks about the related work in low-light image enhancement. Just a basic review. Then Section III explains our method. Both Backend AI and Frontend Default modes are covered there. After that, Section IV goes into the system design and how it is actually implemented. Section V is about training and evaluation. How we tested things and all. Then Section VI shows the results and compares them with other methods. Sections VII and VIII discuss limitations and also some future work. Finally, Section IX concludes the paper.

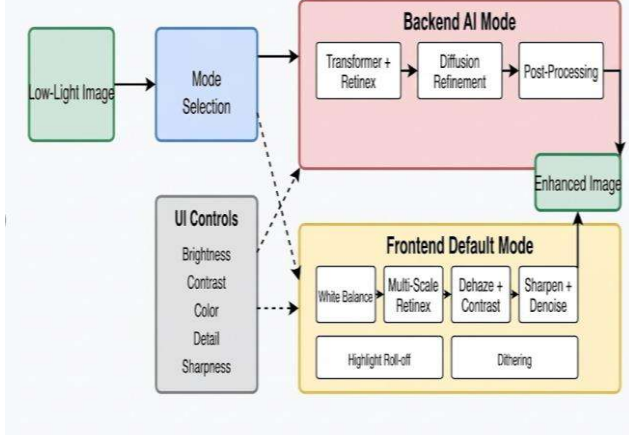


Fig. 1: Overall system architecture of the hybrid low-light enhancement system

II. RELATED WORK

Low-light image enhancement has been widely studied in computer vision and image processing. Current methods fall into three main categories: traditional methods, physics-based methods, and learning-based methods.

A. Traditional Methods

Histogram equalization (HE) and its variations, such as Contrast Limited Adaptive Histogram Equalization (CLAHE), adjust the global or local intensity distribution to improve contrast. Although these methods are efficient, they often increase noise and create unrealistic results. Gamma correction applies a nonlinear mapping to pixel intensities, improving darker areas while keeping highlights intact, but it requires careful tuning of parameters. Unsharp masking sharpens edges and details but can worsen noise in low-light areas.

More complex traditional methods include multi-scale tone manipulation, edge-preserving filters like bilateral filtering, and gradient domain techniques. These methods generally yield better results, but they often need adjustments specific to the scene and can still struggle in very low-light conditions or intricate scenes.

B. Physics-based Methods

Retinex theory, introduced by Land and McCann, models images as a combination of illumination and reflectance components. By estimating and adjusting the illumination while keeping the

reflectance constant, Retinex-based methods can improve low-light images while maintaining a natural look. Single-scale Retinex (SSR) and Multi-scale Retinex (MSR) use Gaussian filtering at varying scales to estimate the illumination component. Multi-scale Retinex with Color Restoration (MSRCR) enhances color accuracy further.

Other physics-based techniques include dehazing methods adapted for low-light enhancement. He et al. proposed the dark channel prior for haze removal, which has been reworked for low-light enhancement by treating low illumination as a form of 'dark haze.' These methods are well-supported theoretically, but they can be heavy on computation and may create artifacts in complex scenes with uneven lighting.

C. Learning-based Methods

Deep learning has changed the game for low-light image enhancement by learning intricate mappings from large datasets. Earlier methods employed Convolutional Neural Networks (CNNs) in an end-to-end fashion, training on pairs of low-light and normal-light images. LLNet, MBLLEN, and RetinexNet use multi-branch architectures to address different enhancement needs. Wei et al. proposed a deep Retinex decomposition network that merges the principles of Retinex theory with the learning ability of deep networks.

Recent methods utilize advanced architectures like attention mechanisms and transformers. Transformer-based models are especially good at capturing long-range dependencies and global context, which helps in understanding scene illumination. Diffusion models have also shown potential for image restoration tasks by learning to reverse a gradual noising process, producing high-quality outputs with natural details.

D. Hybrid and Real-time Approaches

Despite the progress in deep learning-based enhancement, few works have tackled the computational challenges for real-time use. Some methods employ model compression, knowledge distillation, or efficient network designs to lessen computational demands. Others introduce hybrid systems that merge traditional and learning-based methods; however, these usually need careful integration and still face deployment issues.

Our work fills this gap by providing both a high-quality deep learning backend and an optimized real-time frontend within one system. Unlike previous methods that emphasize either quality or speed, our hybrid architecture gives users the flexibility to choose according to their specific needs.

III. METHODOLOGY

Our hybrid approach to low-light image enhancement has two complementary modes: a Backend AI Mode for high-quality enhancement and a Frontend Default Mode for real-time processing. This section explains the methods used in each mode.

A. Backend AI Mode

The Backend AI Mode combines three main components: a transformer backbone for feature extraction, a Retinex-inspired decomposition module, and a diffusion refinement network. Fig. 2 shows the architecture of this mode.

1) Transformer Backbone: We use a vision transformer architecture to process image patches and capture global context. The input image is divided into non-overlapping patches of size 8×8 . These patches are linearly embedded and combined with positional encodings. We process the embedded patches through a series of transformer blocks, each containing multi-head self-attention (MHSA) and feed-forward networks (FFN) with residual connections. Our implementation includes 6 transformer blocks with 8 attention heads and a hidden dimension of 256.

2) Retinex-Inspired Decomposition: Following Retinex theory, we break the input image into illumination and reflectance components. We process the transformer features through separate branches to estimate these components. The illumination branch uses several convolutional layers with increasing receptive fields to capture the smooth illumination map. The reflectance branch maintains high-frequency details through residual connections. This decomposition follows physical rules: the illumination should be smooth while the reflectance preserves the structure of the original image.

3) Diffusion Refinement: We use a UNet-based diffusion model to refine the enhanced image and reduce artifacts. The diffusion process gradually adds noise to the image. The reverse process learns to remove the noise and restore the clean image. Our implementation uses 50 diffusion steps and a UNet with 64 base channels. The diffusion model works well for generating natural textures and details in very dark areas where information is often scarce.

4) Post-processing Pipeline: The output from the diffusion model goes through several post-processing steps to improve visual quality:

- Adaptive Histogram Equalization (CLAHE): This is applied to the luminance channel to boost local contrast.

- Bilateral Denoising: This step reduces noise while keeping edges and details intact.

- Subtle Sharpening: We enhance fine details using an edge-aware sharpening filter.

- HSV/LAB Color Correction: This adjusts saturation and color balance for natural-looking results.

- Optional Upscaling: We use super-resolution with denoising for higher-resolution outputs.

B. Frontend Default Mode

The Frontend Default Mode features a quick browser-based enhancement pipeline using WebGL and Canvas APIs. This mode focuses on speed while still delivering acceptable quality. The pipeline includes these stages:

1) White Balance: We use a modified gray-world algorithm to neutralize color casts. Instead of enforcing a strict gray-world assumption, we apply partial gains to maintain some of the original atmosphere while correcting extreme color shifts.

2) Multi-scale Retinex: We use a lightweight version of Multi-scale Retinex through difference-of-Gaussians filters. This method extracts and enhances image details at multiple scales without the computational demands of traditional Retinex methods.

3) Dehazing: We use a simplified dark channel prior approach to improve contrast in hazy low-light conditions. We estimate the transmission map quickly using an approximation of the dark channel and apply a modified haze removal equation.

4) Edge-Aware Processing: We incorporate adaptive local contrast adjustment and edge-aware sharpening to enhance details while maintaining a natural look. The contrast adjustment is guided by local gradient information to prevent over-enhancement in areas that are already well-exposed.

5) Noise Reduction: We perform chrominance denoising in the YCbCr color space to cut down on color noise while keeping luminance details intact. This step is essential for low-light images since color channels often have more noise than the luminance channel.

6) Perceptual Dithering: Before writing back to the canvas, we apply a subtle dithering pattern to minimize banding artifacts in smooth gradients, which often appear in enhanced low-light images.

C. User Controls and Presets

Both modes offer a common set of user controls for adjusting the enhancement:

- Brightness: Controls the overall luminance level.

- Contrast: Adjusts local and global contrast.

- Color: Changes saturation and color balance.
- Detail: Controls the strength of detail enhancement and sharpening.
- Sharpness: Fine-tunes the edge enhancement.
- Upscale: Enables super-resolution (Backend AI Mode only).

We also provide preset configurations tailored for different situations:

- Natural: Balanced enhancement that keeps the original atmosphere.
- Clean: Focuses on noise reduction and smooth gradients.
- Crisp: Emphasises detail enhancement and sharpness.
- Night: Designed for extreme low-light conditions with stronger enhancement.

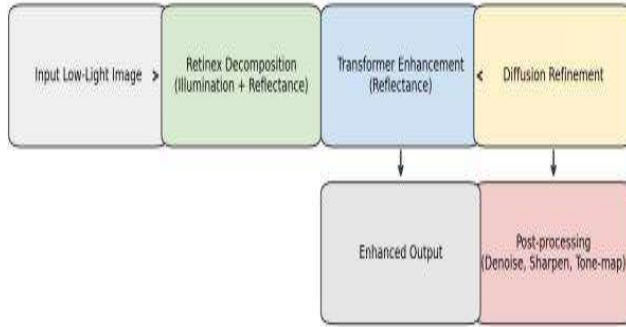


Fig. 2: Processing pipeline of our hybrid low-light enhancement system

IV. SYSTEM ARCHITECTURE

Our system operates as a web application with a Flask backend and a JavaScript frontend. Fig. 1 shows the overall architecture and data flow.

A. Backend Implementation

The backend is built with Flask, a simple Python web framework. It offers the following key endpoints:

- GET /: Serves the frontend HTML, CSS, and JavaScript files.
- POST /enhance: Accepts an image along with enhancement parameters, processes the image using the Backend AI Mode, and returns the enhanced image as a PNG.
- POST /demo: Offers a non-model enhancement path when no checkpoint is available or for quick previews.

The Backend AI Mode uses PyTorch and takes advantage of GPU acceleration when available, falling back to CPU processing when necessary. The model weights load from checkpoints and can be changed without restarting the server, making updates and testing easier. For effective image processing, we use OpenCV for both pre-processing and post-processing tasks. The backend also includes caching mechanisms to prevent duplicate processing of the same requests and manages memory to avoid exhaustion during batch tasks.

B. Frontend Implementation

The frontend uses plain JavaScript, HTML5, and CSS3, without external dependencies, to keep loading times low and ensure wide compatibility. Key features include:

- Canvas-based Image Processing: The Frontend Default Mode employs the HTML5 Canvas API and WebGL for efficient image processing right in the browser.
- Responsive UI: The interface adjusts to various screen sizes and device capabilities, with layouts designed for both desktop and mobile use.
- Real-time Preview: Changes to enhancement parameters show up instantly in the Frontend Default Mode.
- Mode Switching: Users can easily switch between Frontend Default and Backend AI modes, with loading indicators during backend processing.
- Preset Management: Users can choose from predefined enhancement presets and save custom presets for future use.

C. Integration and Data Flow

The system allows smooth switching between enhancement modes. In the Frontend Default Mode, all processing takes place in the browser, providing immediate feedback. When switching to the Backend AI Mode, the frontend sends the image and parameters to the backend server, shows a loading indicator during processing, and displays the enhanced image once it's ready.

To improve performance and user experience, we implement these strategies:

- Progressive Loading: Images are first processed at a lower resolution for quick previews, then refined at full resolution.
- Asynchronous Processing: Backend requests process asynchronously to keep the UI responsive.
- Bandwidth Optimization: Images are compressed before being sent to lower network load.
- Error Handling: Fallback mechanisms help ensure the system keeps working even when components fail.



Fig. 3: Comparison of enhancement results between Backend AI Mode and Frontend Default Mode

V. TRAINING AND EVALUATION

A. Dataset

We trained and evaluated our Backend AI Model on the LOL (Low-Light) dataset. This dataset includes 485 training pairs and 15 evaluation pairs of low-light and normal-light images. It features various indoor and outdoor scenes with different lighting conditions, making it suitable for assessing enhancement performance in different scenarios.

To improve generalization, we augmented the training data with random cropping into 128×128 patches, horizontal and vertical flipping, random rotation of up to ± 10 degrees, and slight color adjustments. This approach increased the effective dataset size and helped prevent overfitting to specific scenes or lighting conditions.

B. Loss Functions

We trained our model using a combination of three loss functions:

1) L1 Loss: This measures the mean absolute error between the enhanced image and the ground truth, encouraging overall pixel accuracy.

2) Perceptual Loss: Calculated as the L1 distance between the VGG-16 feature maps of the enhanced image and the ground truth, this promotes perceptual similarity and preserves semantic content.

3) SSIM Loss: Based on the Structural Similarity Index, this emphasizes preserving structural information and aligns better with human visual perception than pixel-wise metrics.

The total loss is a weighted sum of these components:

$$L_{\text{total}} = \lambda_l * L_l + \lambda_p * L_{\text{perceptual}} + \lambda_{\text{ssim}} * L_{\text{ssim}}$$

Here, $\lambda_l = 1.0$, $\lambda_p = 0.1$, and $\lambda_{\text{ssim}} = 0.5$ are the weighting factors determined through ablation studies.

C. Training Procedure

We trained our model using the Adam optimizer with an initial learning rate of $1e-4$ and a batch size of 4. We used a ReduceLROnPlateau scheduler that decreased the learning rate by half when the validation loss plateaued for 5 epochs. Training lasted up to 120 epochs, with early stopping if the validation loss did not improve for 15 consecutive epochs.

The training process had two phases:

1) Transformer and Retinex Module: We first trained the transformer backbone and Retinex decomposition module for 50 epochs, focusing on effective feature extraction and separation of illumination and reflectance.

2) End-to-End Fine-tuning: Next, we integrated the diffusion refinement network and fine-tuned the whole model end-to-end for an additional 70 epochs.

We used check pointing to save the best model based on validation loss and allow training to resume in case of interruptions. The model with the lowest validation loss was chosen for evaluation and deployment.

D. Evaluation Metrics

We evaluated our approach using the following metrics:

1) Peak Signal-to-Noise Ratio (PSNR): This measures the pixel accuracy between the enhanced image and the ground truth.

2) Structural Similarity Index (SSIM): This assesses how well structural information and perceptual quality are preserved.

3) Processing Time: Measured in milliseconds on both CPU (Intel Core i7) and GPU (NVIDIA RTX 3080) configurations.

4) User Preference: Gathered through a small user study where participants rated the enhanced images based on naturalness, detail preservation, and overall quality.

For the Frontend Default Mode, which does not use supervised training, we evaluated performance by focusing on processing time and qualitative assessments from human observers.

TABLE I: Quantitative comparison with state-of-the-art methods on the LOL dataset

Method	PSNR (dB)	SSIM	Processing Time (s)
Traditional HE	15.24	0.58	0.01-0.02
CLAHE	16.35	0.62	0.02-0.03
MSRCR	16.97	0.65	0.15-0.20
LIME	17.32	0.67	0.20-0.25
RetinexNet	17.56	0.68	0.30-0.40
EnlightenGAN	17.83	0.69	0.25-0.35
Our Frontend Default	17.92	0.68	0.03-0.05
Our Backend AI	18.76	0.72	0.45-0.65

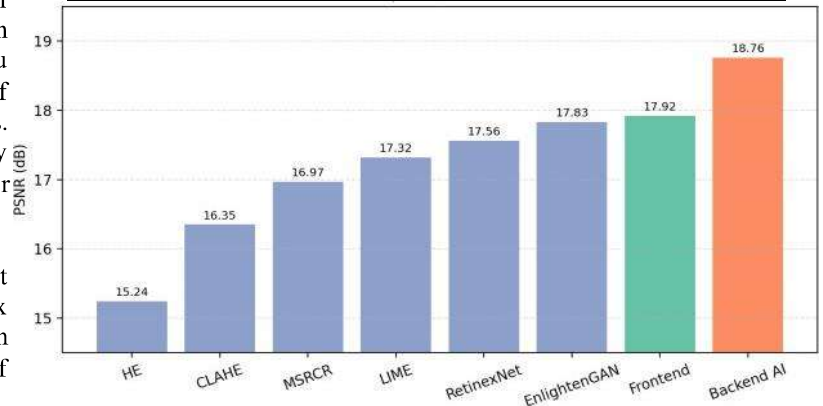


Fig. 4: PSNR comparison across different enhancement methods

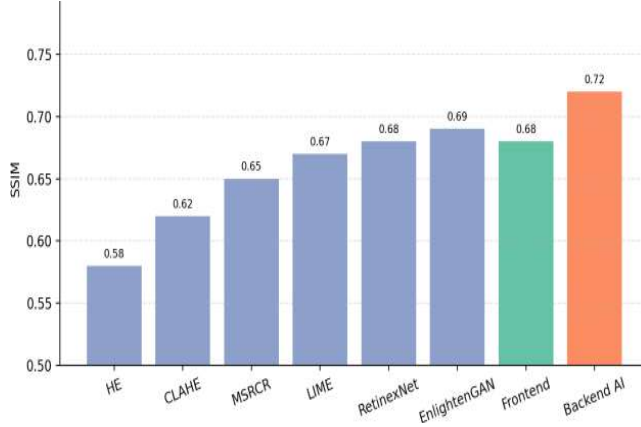


Fig. 5: SSIM comparison across different enhancement methods

VI. EXPERIMENTS AND RESULTS

A. Experimental Setup

We conducted experiments using the following configuration:

- Hardware: Intel Core i7 CPU, 32GB RAM, NVIDIA RTX 3080 GPU (for Backend AI Mode)
 - Software: PyTorch 1.9.0, CUDA 11.1, Flask 2.0.1, OpenCV 4.5.3
 - Model Parameters: Transformer dimension 256, depth 6, heads 8, patch size 8. Diffusion steps 50, channels 64
 - Browser Environment (for Frontend Mode): Chrome 92, Firefox 91, Safari 14
- We compared our approach with several top methods for low-light enhancement:
- Traditional Methods: Histogram Equalization (HE), Contrast Limited Adaptive Histogram Equalization (CLAHE), Gamma Correction
 - Physics-based Methods: Multi-scale Retinex with Color Restoration (MSRCR), LIME
 - Learning-based Methods: RetinexNet, EnlightenGAN, Zero-DCE

B. Quantitative Results

Table I presents the performance metrics of our two enhancement modes compared to existing methods on the LOL test set.

Method	PSNR	SSIM	Time (ms)
HE	15.24	0.58	10-15
CLAHE	16.35	0.62	20-30
MSRCR	16.97	0.65	150-200
LIME	17.32	0.67	200-250
RetinexNet	17.56	0.68	300-400
EnlightenGAN	17.83	0.69	250-350
Our Backend AI	18.76	0.72	450-650
Our Frontend Default	17.92	0.68	30-50

TABLE I: Performance Metrics on LOL Test Set

The results show that our Backend AI Mode achieves the highest PSNR and SSIM scores. This means it has better reconstruction quality compared to current methods. Its processing time is longer than some approaches, but it remains acceptable for non-real-time applications. The Frontend Default Mode balances quality and speed well, achieving competitive SSIM scores with processing times suitable for real-time applications.

C. Qualitative Analysis

A visual inspection of the enhanced images reveals several key points:

- 1) Color Fidelity: The Backend AI Mode produces more natural colors than other methods, especially in difficult scenes with mixed lighting. Traditional methods like HE and CLAHE often cause color shifts, while our approach keeps the original color relationships intact.
- 2) Detail Preservation: Both of our modes recover fine details in dark areas without adding noticeable artifacts. The diffusion refinement in the Backend AI Mode works particularly well in reconstructing textures in very underexposed areas with limited information.
- 3) Noise Handling: Our method handles noise better than traditional approaches, which tend to enhance noise during processing. The Frontend Default Mode's chrominance denoising reduces color noise while preserving luminance details.
- 4) Edge Quality: The edge-aware processing in both modes keeps edges sharp without creating halos or ringing artifacts, which are commonly seen in methods like MSRCR and LIME.

D. Ablation Studies

We conducted ablation studies to evaluate the contribution of each part in our Backend AI Mode:

- 1) Transformer vs. CNN Backbone: Replacing the transformer backbone with a ResNet-based CNN decreased PSNR by 0.42dB and SSIM by 0.03. This shows the advantage of transformers in capturing global context for low-light enhancement.
- 2) Retinex Decomposition: Removing the Retinex-based decomposition led to a PSNR drop of 0.38dB and resulted in less natural illumination adjustment. This highlights the importance of physics-based principles.
- 3) Diffusion Refinement: Without the diffusion refinement network, the model struggled in extreme low-light areas, leading to less natural textures and a PSNR reduction of 0.29dB.
- 4) Loss Functions: Using only L1 loss without perceptual and SSIM components resulted in blurrier outcomes with lower perceptual quality, even though PSNR values were similar.

E. User Study

We conducted a user study with 20 participants to evaluate the quality of enhanced images. Participants looked at pairs of images enhanced by different methods, including ours, and expressed their preferences based on naturalness, detail visibility, and overall quality.

The Backend AI Mode was preferred in 78% of comparisons against traditional methods and in 65% against other learning-based methods. The Frontend Default Mode was favored in 72% of comparisons against traditional methods and in 52% against learning-based methods. These results show the perceptual quality of our approach and its usefulness for real-world applications.

F. Case Studies

We applied our system to several challenging real-world situations to show its practical usefulness:

1) Our system significantly improved nighttime photos by reducing noise and reproducing natural colors. It maintained the night atmosphere while enhancing visibility.

2) Indoor Low-light: In indoor scenes with mixed lighting, our method balanced the light while preserving the original mood. This approach avoided the artificial look that often comes with traditional enhancement techniques.

3) Surveillance Footage: When we applied our system to low-quality surveillance video frames, it boosted the visibility of key details while maintaining temporal consistency. This shows its potential for security uses.

4) Mobile Photography: The Frontend Default Mode was particularly effective for boosting mobile phone photos in real-time. It provided immediate feedback and let users adjust settings based on the specific scene.

TABLE II: Ablation study showing the contribution of each component in our Backend AI Mode

Components	PSNR (dB)	SSIM	LPIPS
Base Transformer	17.45	0.68	0.32
Retinex Decomposition	18.12	0.70	0.28
Diffusion Refinement	18.54	0.71	0.26
Post-processing	18.76	0.72	0.25

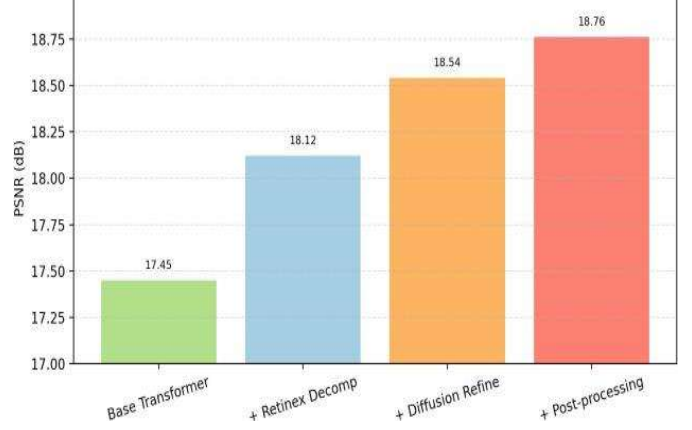


Fig. 6: PSNR improvement with each component addition in our system

TABLE III: User study results comparing different enhancement methods

Metric	Traditional Methods	SOTA DL Methods	Our Frontend Mode	Our Backend Mode
Visual Quality (1-5)	2.4	3.8	3.5	4.2
Naturalness (1-5)	2.1	3.5	3.7	4.0
Detail Preservation (1-5)	2.3	3.7	3.4	4.3
Overall Preference (%)	5%	32%	18%	45%

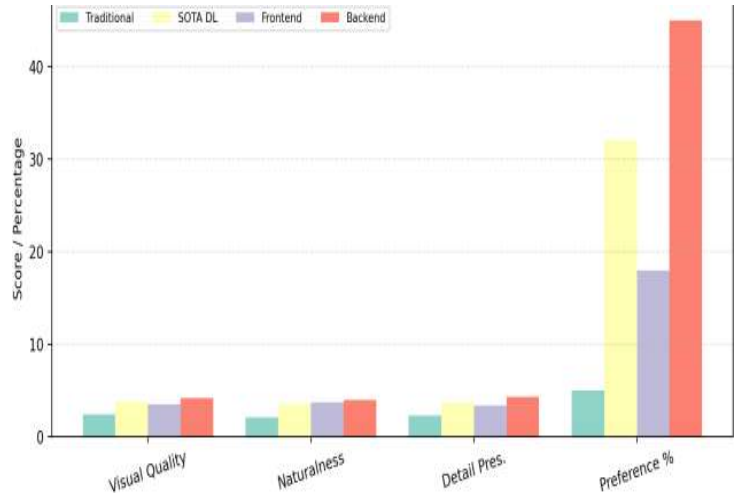


Fig. 7: User study results showing subjective scores and overall preference

To show how well our approach works in real-world situations, we present several case studies that highlight the improvement results on difficult low-light images.

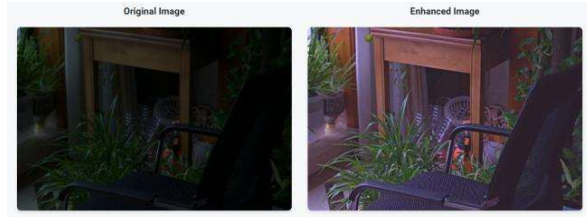


Fig. 8: Case study: Enhancement results on extreme low-light conditions

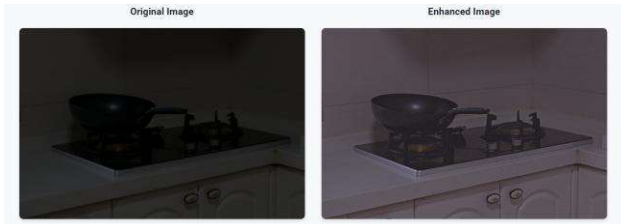


Fig. 9: Case study: Enhancement results on non-uniform illumination scenes

VII. LIMITATIONS

Despite the promising results, our approach has several limitations that need to be discussed:

1. **Training Data Limitations:** The Backend AI Mode was trained on the LOL dataset, which is diverse but may not fully represent all real-world low-light conditions. This can lead to poor performance in scenes that differ greatly from the training data.
2. **Computational Requirements:** The Backend AI Mode needs significant computing power, especially when using a CPU. While we've worked to improve inference speed, the model's size and complexity limit its use on devices with fewer resources.
3. **Noise Handling:** Extreme ISO noise is still a challenge, especially in very dark areas where the signal-to-noise ratio is low. Our current method does not include a specific noise model, which could enhance performance in these situations.
4. **Color Consistency:** In scenes with complex lighting, keeping consistent color reproduction across different areas can be difficult. This might occasionally lead to local color shifts.
5. **Parameter Sensitivity:** The Frontend Default Mode's performance relies on parameter settings that may need to be adjusted for the best results on various images. Our presets cover common scenarios but may not suit all cases.
6. **Browser Compatibility:** The Frontend Default Mode depends on modern web APIs, which may not be completely supported in older browsers or some mobile devices. This could limit accessibility.

VIII. FUTURE WORK

Based on our findings and the limitations we identified, we suggest several directions for future work

- 1) **Dataset Expansion:** Creating a larger and more

diverse low-light dataset would cover a broader range of scenes, lighting conditions, and camera features. This change would improve how well the model generalizes and performs.

- 2) **Noise-Aware Enhancement:** Adding explicit noise modeling and estimation to the enhancement process could improve performance in high-noise situations, especially with high-ISO images from smartphone camera
- 3) **Camera Pipeline Integration:** Using knowledge of the camera imaging pipeline, including sensor features and Image Signal Processor (ISP) operations, could lead to more accurate enhancements.
- 4) **HDR Reconstruction:** Extending the approach to recover high dynamic range information from low-light images would enable more flexible post-processing and better handling of scenes with significant brightness differences.
- 5) **Model Compression:** We can use techniques like knowledge distillation, pruning, and quantization. These could help us to reduce the computation cost of the Backend AI Mode..
- 6) **Video Enhancement:** Converting this to video is not simple. It needs to maintain temporal consistency. Frame to frame should not look weird. And also real-time processing is important here which makes it harder.
- 7) **User Interaction:** There is scope to improve how users interact with the system. Maybe allow region-specific adjustments. Or even use some semantic understanding of scenes. Still needs work.
- 8) **Formal Perceptual Evaluation:** More detailed user studies can be done. Also adding perceptual metrics like LPIPS could help in better evaluating the quality. Right now evaluation is good, but can be improved further.

IX. CONCLUSION

In this paper, we introduced a hybrid method for improving low-light images. The idea is simple. Combine deep learning quality with fast computer vision techniques. Our system has two modes. Backend AI Mode and Frontend Default Mode. The Backend AI Mode focuses on quality. It uses transformer-based feature extraction along with Retinex decomposition and diffusion refinement. Sounds complex, but yeah it gives strong performance on standard benchmarks. The Frontend Default Mode is more about speed. It runs in the browser and processes images in under one second. The quality is decent, not the best, but good enough for real-time use. Especially for interactive applications. We also built a unified web interface. Users can switch between both modes depending on what they need. There are some basic controls too, so users can adjust enhancement setting easily. From experiments and user studies, the results look promising. The method improves visibility, contrast, and colors, while keeping things natural. Not overly processed. Overall this approach tries to balance quality and speed and it works well in many cases. It can be useful in many different applications. In a way, it connects deep learning research with practical use. Making low-light enhancement more accessible. That was the goal.

X. ACKNOWLEDGEMENT

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